IGCSE 0607 Practice Questions — E6.5

Compound shapes and parts of shapes

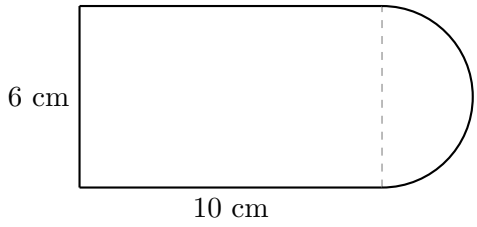
Total Questions: 36 — **Total Marks:** 267 Marks

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- **E6.5** Compound shapes and parts of shapes
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(23 Qs)*

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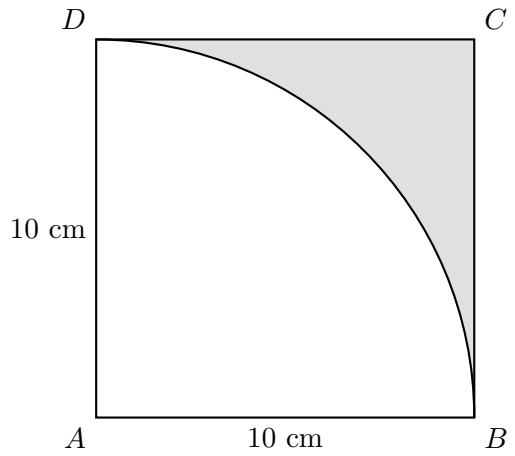
1 A shape is made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
The semicircle is attached to one of the 6 cm sides.



Find the exact perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

2 The diagram shows a square $ABCD$ with side length 10 cm.
A quarter circle with center A and radius 10 cm is drawn inside the square.
The region inside the square but outside the quarter circle is shaded.



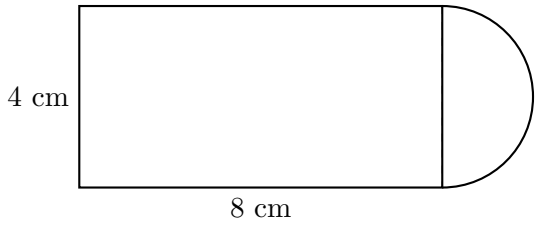
(a) Find the exact perimeter of the shaded region.
Give your answer in terms of π.

..... cm **[2]**

(b) Find the exact area of the shaded region.
Give your answer in terms of π.

..... cm^2 **[2]**

3 The diagram shows a compound shape formed by joining a semi-circle to a rectangle.
The rectangle has a length of 8 cm and a width of 4 cm.
The diameter of the semi-circle is 4 cm.



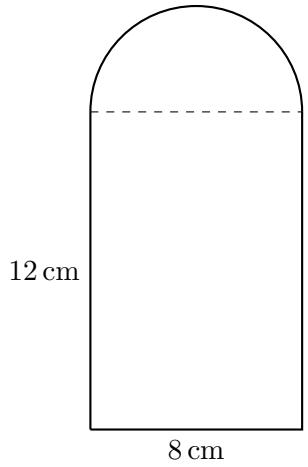
(a) Calculate the perimeter of the shape.
Give your answer in its simplest form, in terms of π.

..... *cm* **[2]**

(b) Calculate the area of the shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

4 A compound shape is made from a rectangle and a semicircle.
The rectangle has width 8 cm and length 12 cm. The semicircle is attached to one of the sides of width 8 cm.



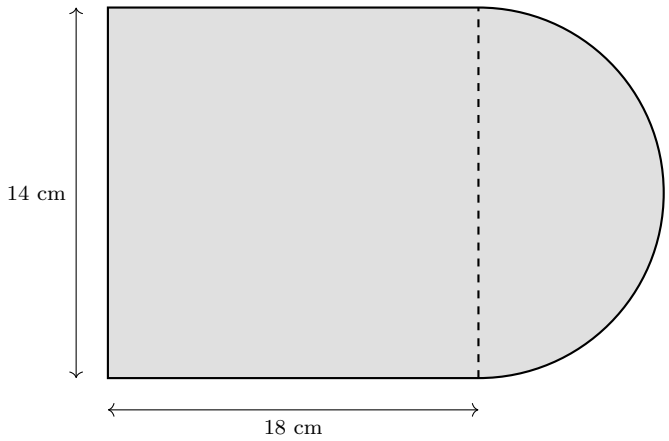
(a) Calculate the exact perimeter of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm **[2]**

(b) Calculate the exact area of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

5 The diagram shows a compound shape made by attaching a semicircle of diameter 14 cm to one end of a rectangle of length 18 cm and width 14 cm.



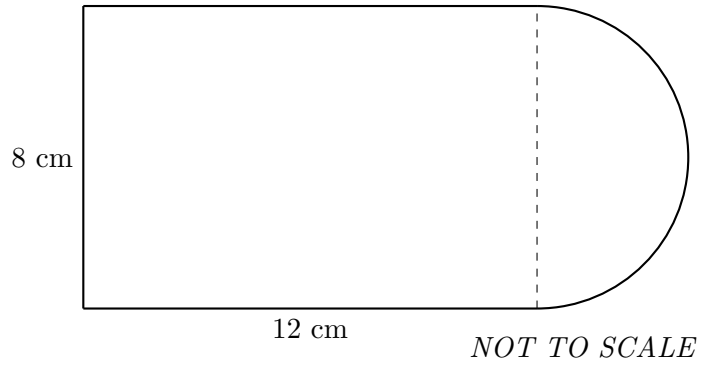
(a) Calculate the perimeter of the compound shape.
Give your answer in terms of π.

Perimeter = ... cm **[2]**

(b) Calculate the total area of the compound shape.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

6 The diagram shows a shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 8 cm.
The semicircle has a diameter of 8 cm.



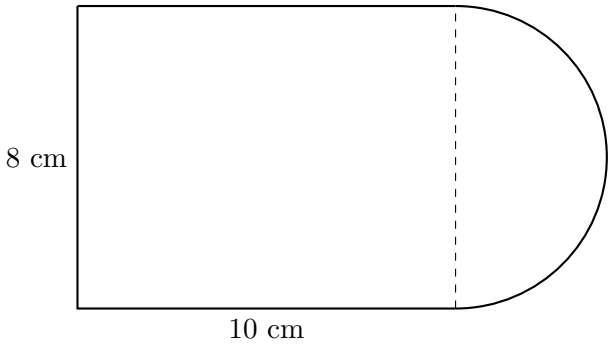
(a) Find the perimeter of the shape.
Give your answer in its simplest form in terms of π.

Perimeter = ... cm **[2]**

(b) Find the area of the shape.
Give your answer in its simplest form in terms of π.

Area = ... cm^2 **[3]**

7 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 8 cm.
The semicircle is attached to one of the 8 cm sides.



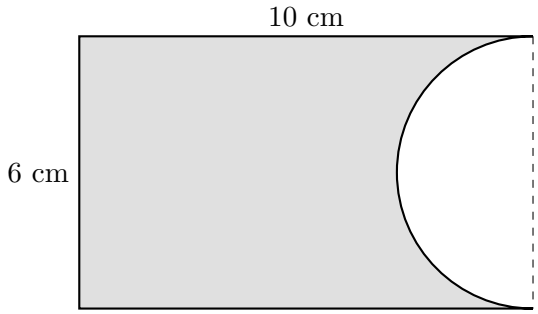
(a) Work out the exact area of the shape.
Give your answer in the form $a + b\pi$, where a and b are integers.

Area = ... cm^2 **[3]**

(b) Work out the exact perimeter of the shape.
Give your answer in the form $c + d\pi$, where c and d are integers.

Perimeter = ... cm **[3]**

8 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
A semicircle of diameter 6 cm has been removed from one end of the rectangle.



NOT TO SCALE

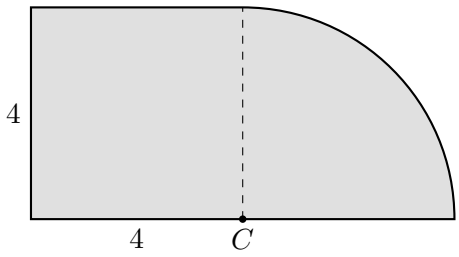
(a) Work out the perimeter of the shape.
Give your answer in the form $a + b\pi$, where a and b are integers.

..... cm **[3]**

(b) Work out the area of the shape.
Give your answer in the form $c - d\pi$, where c is an integer and d is a decimal.

..... cm^2 **[3]**

9 The diagram shows a shape formed by a square of side 4 cm and a quarter of a circle with center C.



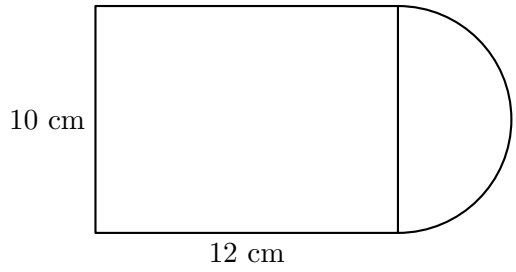
(a) Find the area of the shape.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

(b) Find the perimeter of the shape.
Give your answer in terms of π.

Perimeter = ... cm **[3]**

10 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 10 cm.
The semicircle has a diameter of 10 cm.



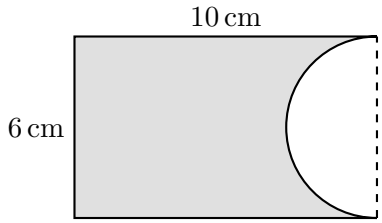
(a) Calculate the perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

(b) Calculate the area of the shape.
Give your answer in the form $c + d\pi$.

..... cm^2 **[3]**

11 The diagram shows a shape made from a rectangle with dimensions 10 cm by 6 cm. A semicircle of diameter 6 cm has been removed from one end.



NOT TO SCALE

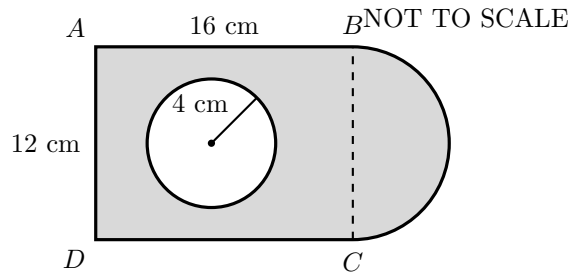
(a) Calculate the area of the shape.
Give your answer in terms of π.

..... cm^2 **[3]**

(b) Calculate the perimeter of the shape.
Give your answer in terms of π.

..... cm **[3]**

12 The diagram shows a shaded metal plate formed from a rectangle $ABCD$ with a semicircle of diameter BC attached to side BC.
Inside the rectangle, a circular hole of radius 4 cm is cut out.
$AB = 16$ cm and $BC = 12$ cm.



(a) Giving your answers in terms of π, calculate

(i) the outer perimeter of the metal plate,

..... cm **[2]**

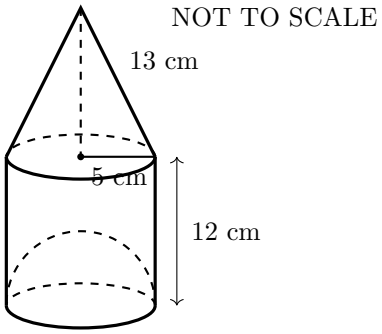
(ii) the area of the shaded region.

..... cm^2 **[3]**

(b) The shaded metal plate is the uniform cross-section of a solid prism of thickness 5 cm. Calculate the volume of the prism, giving your answer in the form $p + q\pi$, where p and q are integers.

..... cm^3 **[2]**

13 The diagram shows a compound solid formed from a right circular cylinder of base radius 5 cm and height 12 cm. A right circular cone of base radius 5 cm and slant height 13 cm is joined to the top face of the cylinder, and a hemisphere of radius 5 cm is hollowed out of the bottom face of the cylinder.



(a) Show that the vertical height of the conical part is 12 cm.

[2]

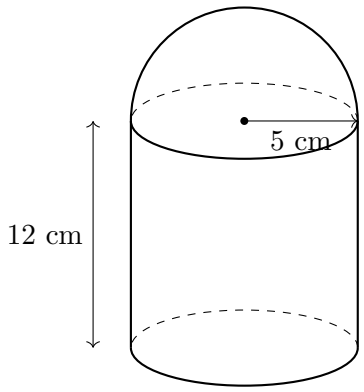
(b) Calculate the total surface area of the compound solid, giving your answer in terms of π.

..... cm^2 **[4]**

(c) Calculate the volume of the compound solid, giving your answer as a single fraction in terms of π.

..... cm^3 **[4]**

14 A solid is made by attaching a hemisphere of radius 5 cm to one end of a solid cylinder.
The cylinder has a radius of 5 cm and a height of 12 cm.



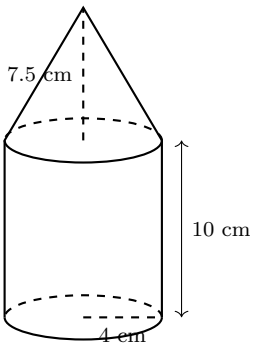
(a) Calculate the total volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[3]**

15 A solid metal trophy consists of a right circular cone placed on top of a cylinder, both of base radius 4 cm.
The cylinder has height 10 cm and the cone has perpendicular height 7.5 cm.



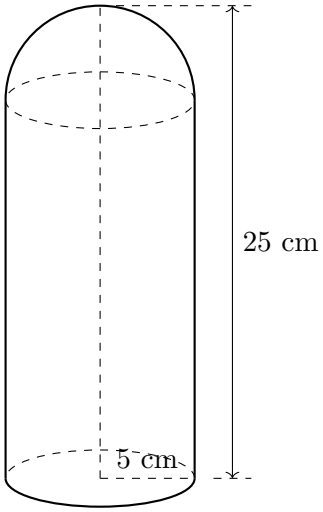
(a) Calculate the total volume of the solid trophy.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid trophy.

Surface area = ... cm^2 **[3]**

16 The diagram shows a solid in the shape of a cylinder with a hemisphere attached to the top.
The radius of both the cylinder and the hemisphere is 5 cm.
The total height of the solid is 25 cm.



NOT TO SCALE

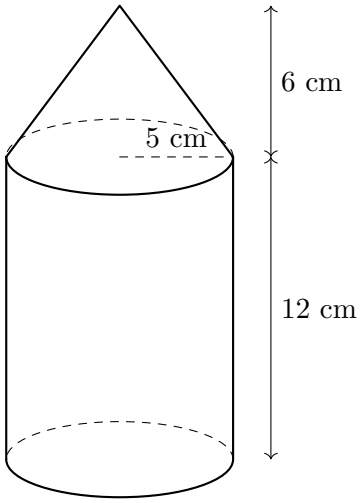
(a) Calculate the total volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

17 The diagram shows a solid compound shape made from a cylinder and a cone.
The cylinder and the cone both have a base radius of 5 cm.
The height of the cylinder is 12 cm and the height of the cone is 6 cm.



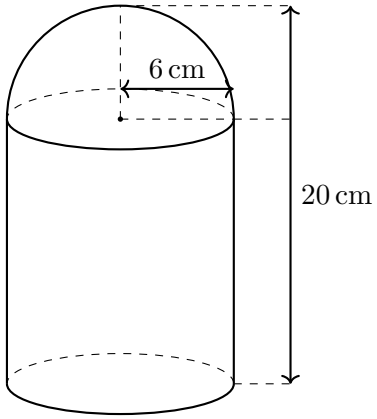
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

18 A solid is made by placing a solid hemisphere on top of a solid cylinder.
The cylinder and the hemisphere both have radius 6 cm.
The total height of the solid is 20 cm.



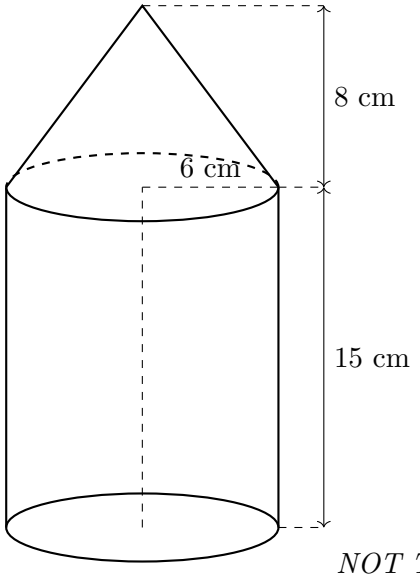
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

19 A solid is made from a cylinder and a cone.
The cylinder has radius 6 cm and height 15 cm.
The cone has radius 6 cm and height 8 cm.



NOT TO SCALE

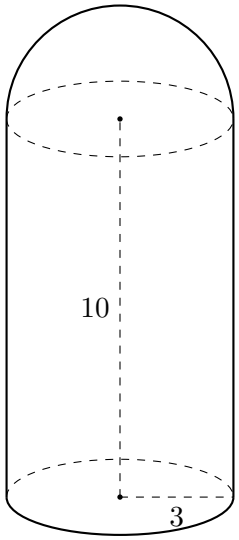
(a) Calculate the total volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

20 A solid is made by attaching a hemisphere of radius 3 cm to one end of a solid cylinder of radius 3 cm and height 10 cm.



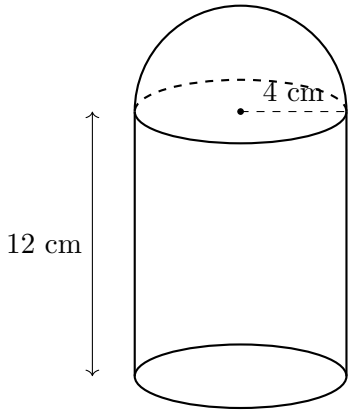
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Total surface area = ... cm^2 **[4]**

21 A solid object is formed by joining a hemisphere to one end of a solid cylinder.
The cylinder has a radius of 4 cm and a length of 12 cm.
The hemisphere has the same radius as the cylinder.



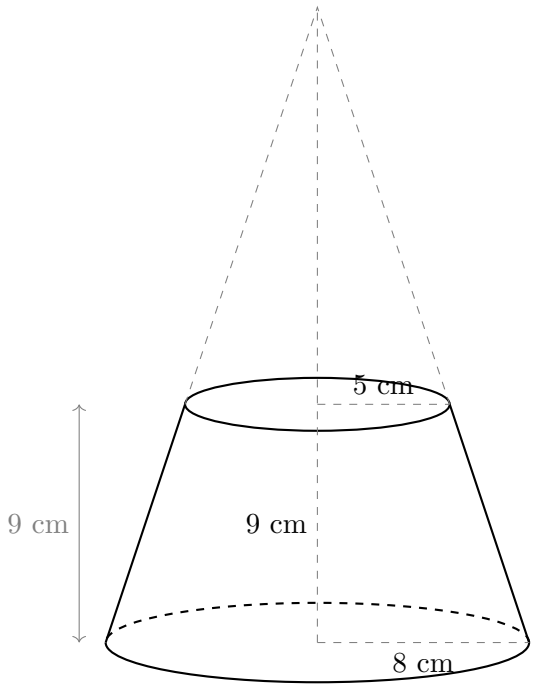
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

22 The diagram shows a solid frustum.
The frustum is made by removing a smaller cone from a larger cone.
The larger cone has a base radius of 8 cm and a vertical height of 24 cm.
The smaller cone has a base radius of 5 cm and a vertical height of 15 cm.



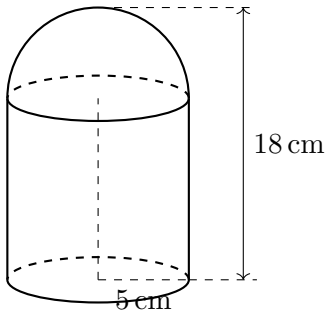
(a) Calculate the volume of the solid frustum.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[5]**

23 A solid is formed by placing a hemisphere of radius 5 cm on top of a cylinder of radius 5 cm.
The total height of the solid is 18 cm.



NOT TO SCALE

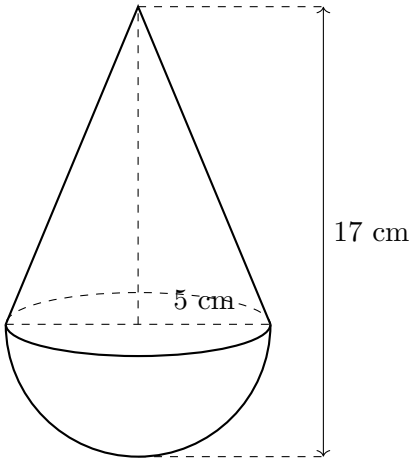
(a) Calculate the volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[4]**

24 The diagram shows a solid made by joining a solid hemisphere to a solid cone.
The common radius of the hemisphere and the base of the cone is 5 cm.
The total height of the solid is 17 cm.



(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

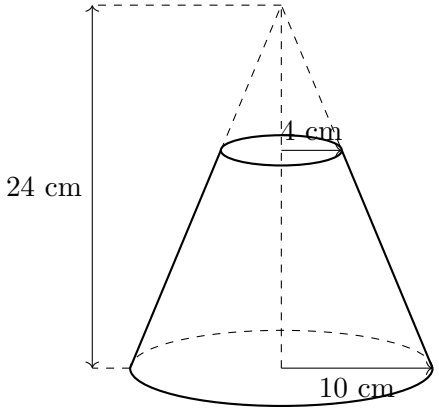
(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[3]**

(c) The solid is melted down and completely recast into a single solid cylinder of radius 4 cm.
Calculate the height of the cylinder.

Height = ... cm **[3]**

25 A solid cone of height 24 cm and base radius 10 cm is cut parallel to its base to form a frustum and a smaller cone.
The radius of the top circular face of the frustum is 4 cm.



(a) Show that the height of the removed small cone is 9.6 cm.

[2]

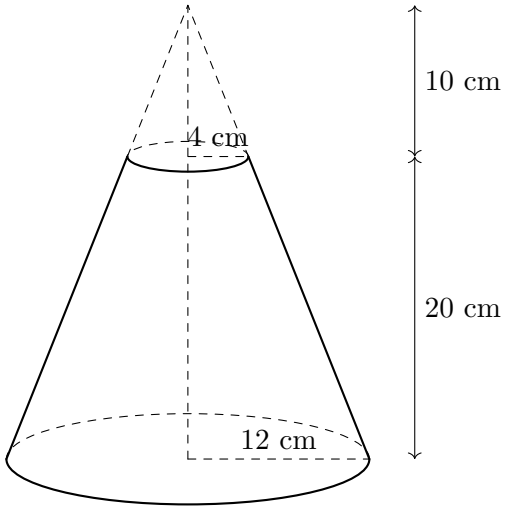
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[3]**

26 The diagram shows a solid metal frustum.
The frustum is formed by removing a small cone of height 10 cm from a larger cone.
The radius of the circular base of the frustum is 12 cm and the radius of the circular top is 4 cm.
The height of the frustum is 20 cm.



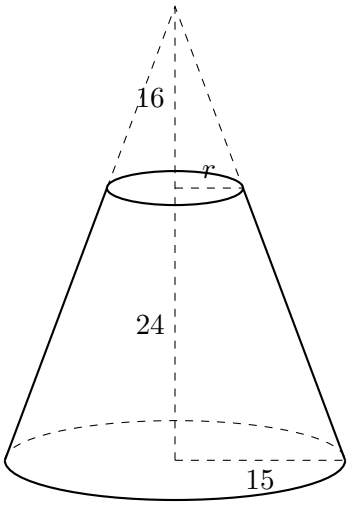
(a) Calculate the volume of the solid frustum.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

27 A container is in the shape of a frustum of a cone.
The original cone has a base radius of 15 cm and a vertical height of 40 cm.
A smaller cone of vertical height 16 cm is removed from the top to form the frustum.



(a) Show that the radius r of the top circular face of the frustum is 6 cm.

[2]

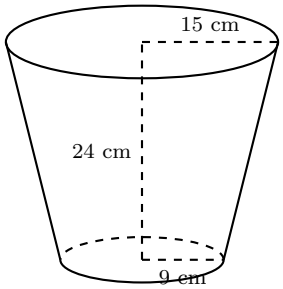
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of the sphere.

Radius = ... cm **[3]**

28 The diagram shows an open bucket in the shape of a frustum of a right circular cone.
The circular base has radius 9 cm and the open circular top has radius 15 cm.
The perpendicular height of the bucket is 24 cm.



(a) Show that the perpendicular height of the original cone before the bottom was removed is 60 cm.

[2]

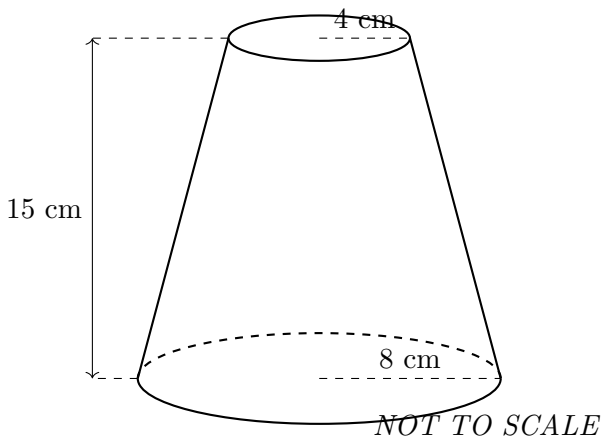
(b) Calculate the capacity of the bucket in litres.

Capacity = ... litres **[3]**

(c) Calculate the total external surface area of the open bucket.

Surface area = ... cm^2 **[3]**

29 A solid frustum of a cone has top radius 4 cm, bottom radius 8 cm and vertical height 15 cm.
It is formed by removing a smaller cone from the top of a larger cone.



(a) Show that the original height of the cone before the top was removed was 30 cm.

[2]

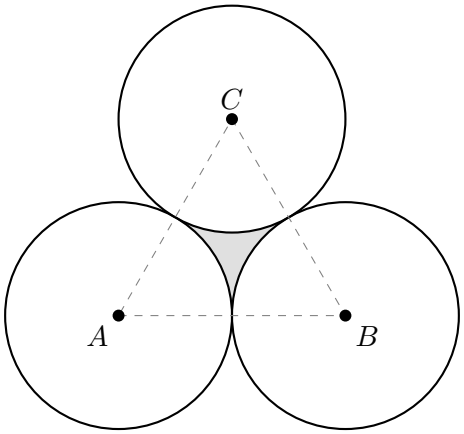
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the solid frustum.

Surface area = ... cm^2 **[4]**

30 The diagram shows three identical circles, each of radius r cm.
The circles touch each other externally.
The region enclosed between the three circles is shaded.



(a) Show that the area of the shaded region, A, is given by the formula

$$A = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$$

[4]

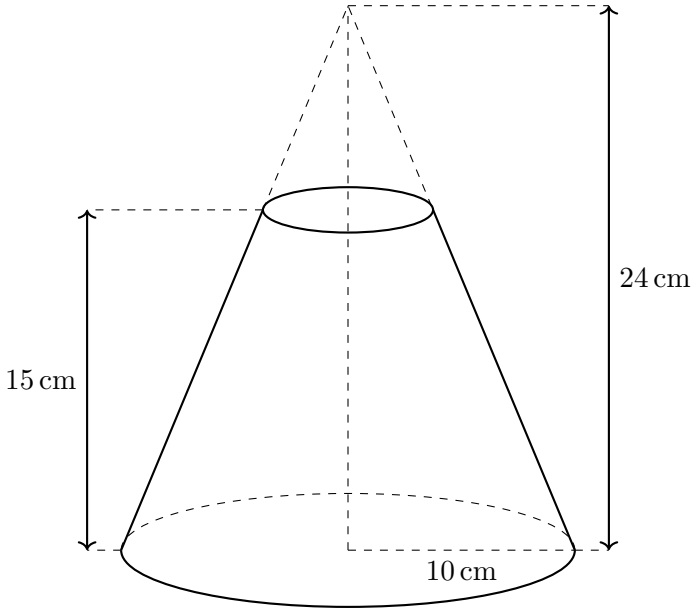
(b) The area of the shaded region is 15 cm^2.
Work out the value of r.

$r =$ ..... **[3]**

(c) Calculate the total perimeter of the shaded region.

..... cm **[3]**

31 A frustum is made by removing a smaller cone from the top of a larger cone.
The larger cone has base radius 10 cm and vertical height 24 cm.
The frustum has vertical height 15 cm.



(a) Find the radius of the top of the frustum.

..... cm **[2]**

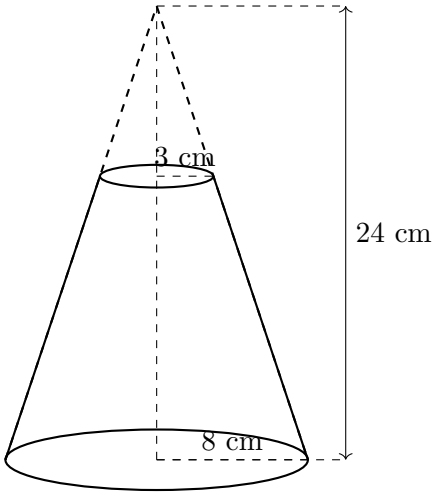
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

..... cm^2 **[5]**

32 A frustum is made by removing a small cone from the top of a large cone.
The large cone has a base radius of 8 cm and a total height of 24 cm.
The removed small cone has a base radius of 3 cm.



(a) Show that the height of the small cone removed is 9 cm.

[2]

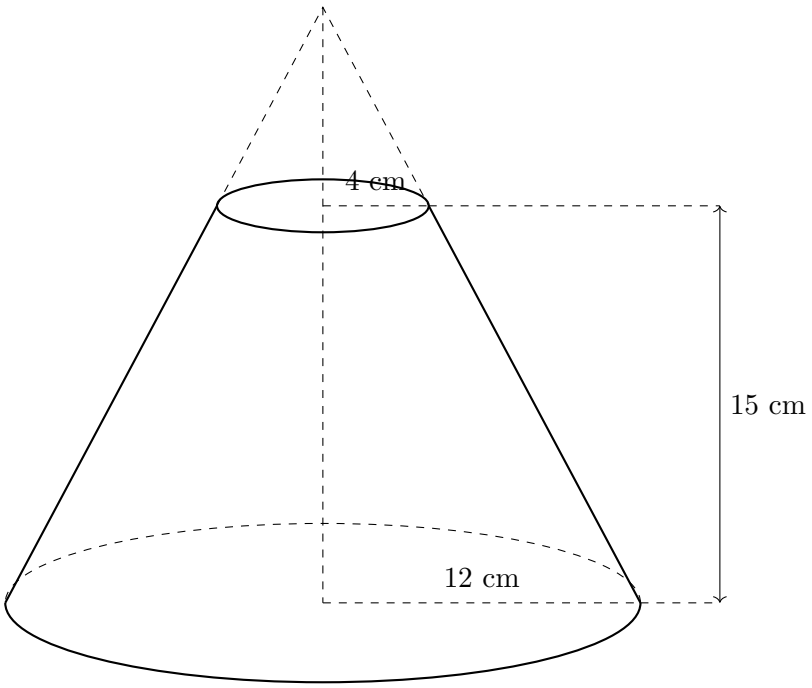
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

Surface area = ... cm^2 **[5]**

33 The diagram shows a solid frustum of a cone.
The frustum is made by removing a small cone from the top of a larger solid cone.
The base radius of the frustum is 12 cm and the top radius is 4 cm.
The height of the frustum is 15 cm.



(a) Calculate the height of the small cone that was removed.

Height = ... cm **[3]**

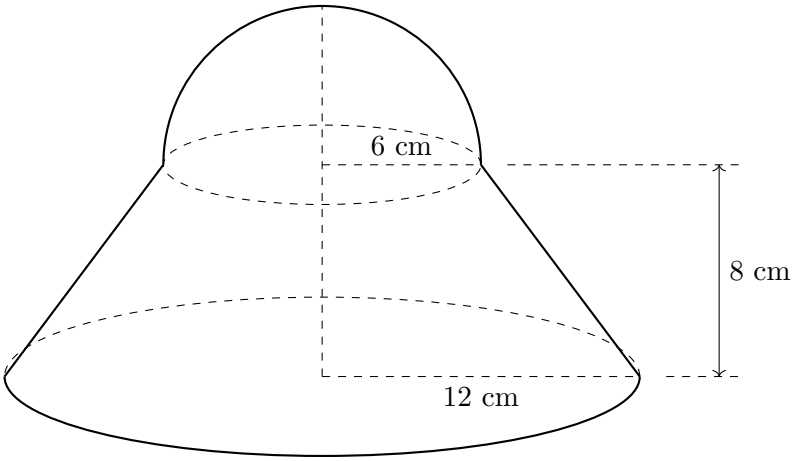
(b) Calculate the volume of the solid frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the solid frustum.

Surface area = ... cm^2 **[5]**

34 The diagram shows a compound solid made from a frustum of a cone and a solid hemisphere.
The frustum is created by removing a small cone from a large cone.
The base radius of the frustum is 12 cm and its top radius is 6 cm.
The height of the frustum is 8 cm.
A solid hemisphere of radius 6 cm is attached to the top face of the frustum.



NOT TO SCALE

(a) Show that the perpendicular height of the large cone from which the frustum is made is 16 cm.

[3]

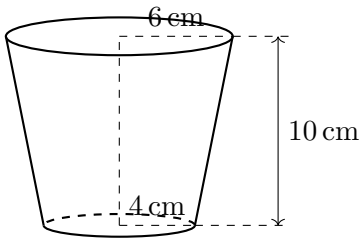
(b) Calculate the total volume of the compound solid.

..... cm^3 **[4]**

(c) Calculate the total surface area of the compound solid.

..... cm^2 **[4]**

35 The diagram shows a closed container in the shape of a frustum of a cone.
The radius of the top circular face is 6 cm and the radius of the base is 4 cm.
The vertical height of the container is 10 cm.



NOT TO SCALE

(a) The frustum is made by removing a smaller cone from a larger cone.
Calculate the vertical height of the larger cone.

..... cm **[3]**

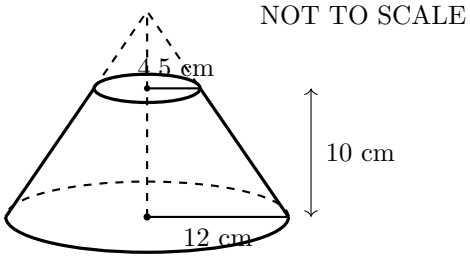
(b) Calculate the volume of the container.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(c) Calculate the total surface area of the closed container.
Give your answer correct to 3 significant figures.

..... cm^2 **[5]**

36 The diagram shows a solid frustum formed by removing a small right circular cone of base radius 4.5 cm from the top of a larger right circular cone of base radius 12 cm. The vertical height of the frustum is 10 cm.



(a) Use similar triangles to show that the vertical height of the removed cone is 6 cm.

[2]

(b) Calculate the total surface area of the solid frustum, giving your answer correct to 3 significant figures.

..... cm^2 **[4]**

(c) Calculate the volume of the solid frustum, giving your answer correct to 3 significant figures.

..... cm^3 **[3]**

(d) The solid frustum is melted down and recast into n identical solid spheres, each of radius 2.4 cm. Calculate the greatest possible integer value of n.

$n =$ ..... **[3]**